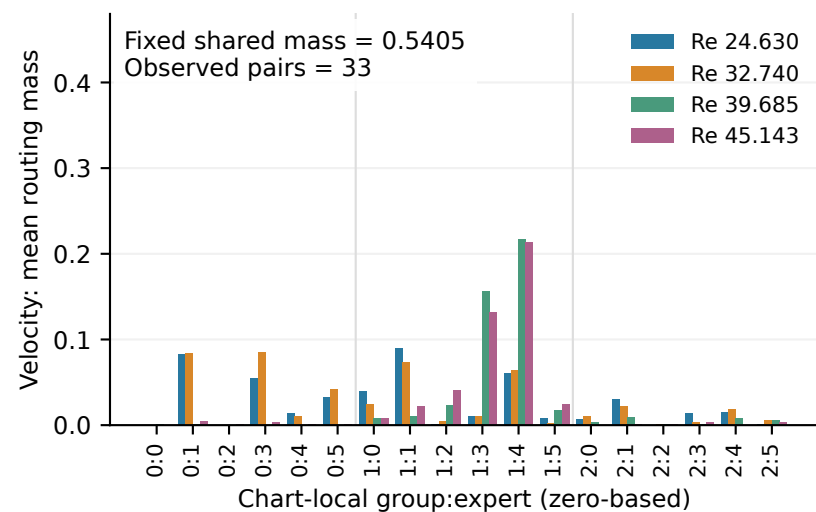
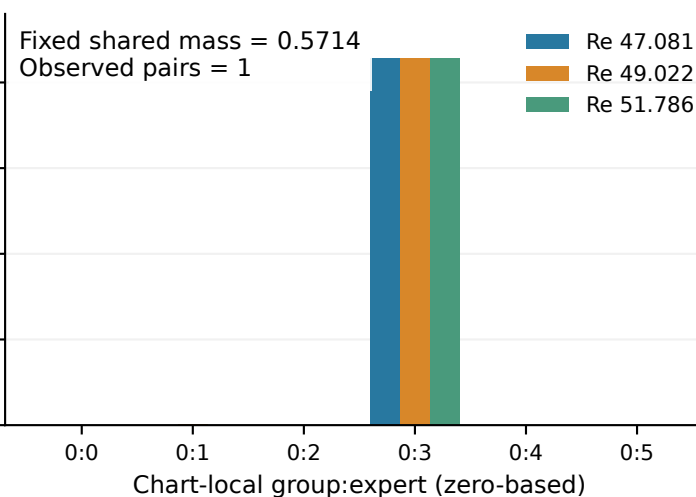


Circular-cylinder internal routing on held-out rollout windows

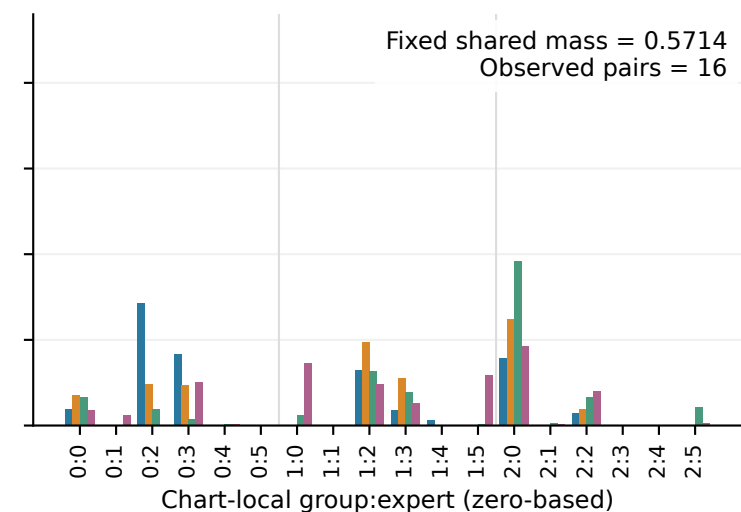
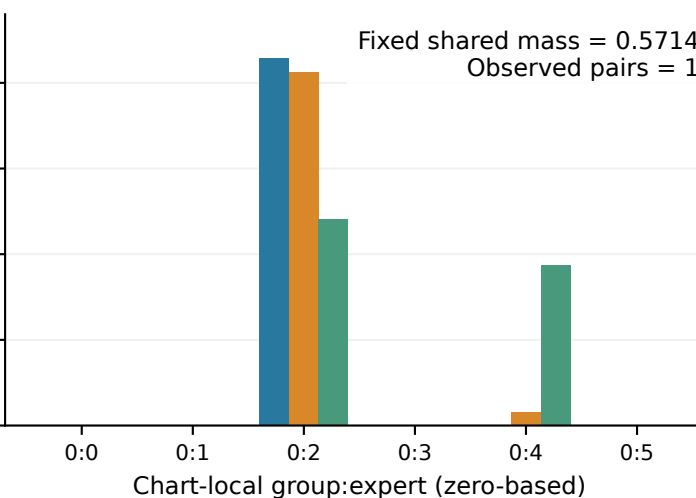
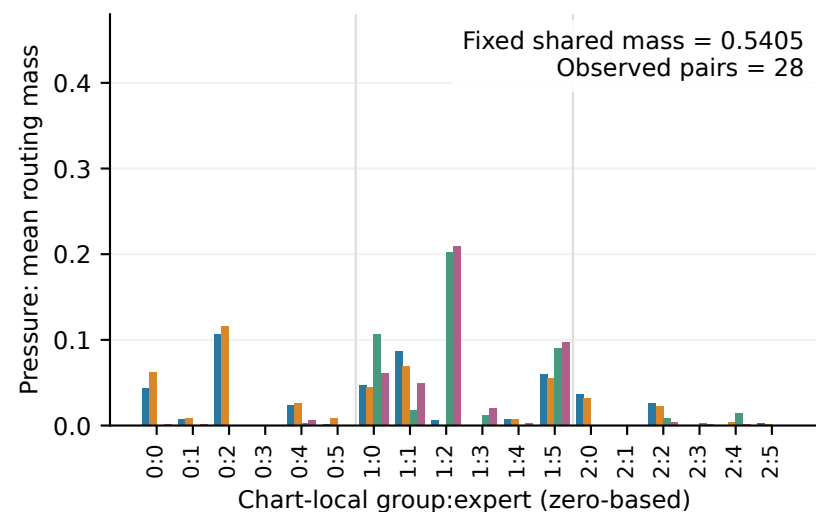
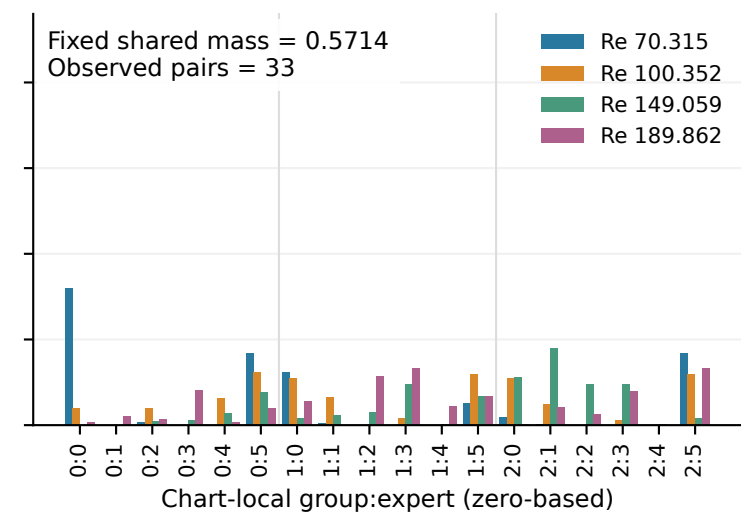
Steady S4 (independent diagnostic)
g0: 22.8% / g1: 68.5% / g2: 8.8%



Hopf H4
g0: 100.0%



Periodic epoch 85
g0: 30.4% / g1: 33.1% / g2: 36.5%



One sample per macro-step (stage 1). All zero-use experts retained. Hopf has one group by design and fixed selected pairs.
S4 is not the current paper Steady checkpoint. Shared mass is a fixed coefficient, not measured output contribution.